

The title

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Abstract

One or two sentences providing a **basic introduction** to the field, comprehensible to a scientist in any discipline.

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The title

Introduction

Sleep is an important determinant of population health [1]. The consequences of poor sleep are profound and include increased all-cause mortality [2], cardiovascular disease [3], cancers [4], and poor mental health [5]. Poor sleep is common: over 50% of children and adolescents [6] and 35% of adults [7] report less sleep than recommended. Improving sleep should be a priority for public health policy and more safe and accessible strategies that promote sleep are needed [8]. In adults, being physically active appears to improve both the quantity and quality of sleep [9] and the World Health Organisation recommends physical activity as a method to improve sleep in adults [10]. In children, adolescents, and older adults associations are unclear [8,11,12] and guidelines do not currently support a role for physical activity to improve sleep.

Promoting physical activity may be an effective method to improve sleep across the lifespan but the published literature to date has at least two limitations that prevent firm conclusions to support health guidelines. First, most studies have relied on self-report measures of sleep and physical activity [8,11,12], which could confound associations and limit the precision of analyses designed to identify a dose-response relationship between physical activity and sleep. Second, no study has collected data from a large sample that included participants across the lifespan. Umbrella reviews [8] and meta-analyses [3,4,9,11–28] have provided some suggestive evidence about how physical activity as a means of improving sleep may vary across the lifespan. However, their reliance on aggregated study-level effects limits the quality and scope of data that can be used to analyse the influence of age [29]. Individual participant data (IPD) analysis can overcome these limitations [30].

In this study we harmonised individual participant data (IPD) from activity monitor observations obtained from studies around the world. The objective of this pooled IPD

study was to investigate the relationship between device-measured physical activity and sleep across the lifespan. We examined whether individuals who are more physically active have better sleep, whether age, sex, and socio-cultural factors moderate the association between physical activity and sleep, and whether there is a bidirectional association between physical activity and sleep.

Results

The aggregated data-set describes 220,612 observations of daily physical activity and sleep from 28,823 unique participants. Table 1 shows demographic information for all participants. Observations were not uniform across the days of the week $\chi^2_{(6)} = 80.75$, $p < .001$. There were fewer observations on Monday ($z = -7.80$) and Sunday ($z = -3.01$); and more on Wednesday ($z = 2.98$). A table of study characteristics can be found in supplementary materials.

The effects of physical activity volume on sleep

We estimated the effects of physical activity on sleep (RQ1) using mixed-effects models. We estimated the effect of physical activity volume on sleep by age, and the results are presented in Table 2 and Figure 1. There was no meaningful relationship between physical activity volume and sleep duration or sleep efficiency. Physical activity volume and sleep onset had a very weak positive association for younger individuals, but a negative association for older individuals, where sleep onset was reduced among those with the highest physical activity. There was a strong positive association between physical activity volume and sleep regularity, which was strongest among older participants. For participants aged 35 years and above, this link weakened among those with a physical activity volume greater than two standard deviations.

The effects of physical activity intensity on sleep

We estimated how physical activity intensity affects sleep across different age groups. We present the results controlling for sex, SES, and BMI, in Table 2 and Figure 2. We found no evidence that physical activity intensity is related to sleep duration, sleep efficiency, or sleep onset. However, we did find evidence for an association between physical activity intensity and improved sleep regularity, which weakened as age increased.

The effects of sleep duration on physical activity

We estimated the effect of sleep duration on physical activity by age. Results, controlling for sex, SES, and BMI are presented in Table 3 and Figure 3. As age increases, both physical activity volume and intensity decrease. We found no evidence for an association between average sleep duration and physical activity volume.

The effects of sleep efficiency on physical activity

We estimated the effect of sleep efficiency on physical activity by age. Results, controlling for sex, SES, and BMI are presented in Table 3 and Figure 4. We did not find a relationship between physical activity volume and sleep efficiency. However, there was a weak linear association between sleep efficiency and more intense physical activity.

The effects of sleep onset on physical activity

We estimated the effect of sleep onset on physical activity by age. Results, controlling for sex, SES, and BMI are presented in Table 3 and Figure 5. There were weak curvilinear relationships where average sleep onset was linked to the highest levels of physical activity volume and intensity. The U-shaped relationship between sleep onset and physical activity volume attenuated for older participants.

The effects of sleep regularity on physical activity

We estimated the effect of sleep regularity on physical activity by age. Results, controlling for sex, SES, and BMI are presented in Table 3 and Figure 6. There was a U-shaped relationship between sleep regularity and physical activity volume. Participants with below-average sleep regularity tended to have average physical activity volume. Increases in regularity above the average were linked to greater physical activity volume. There was a linear relationship between sleep regularity and physical activity intensity which slightly attenuated with age. Greater sleep regularity was associated with greater physical activity the following day.

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Table 1

Participant characteristics

Characteristic	Age group				
	2-11 years	12-18 years	19-35 years	36-65 years	66+ years
Numeric variables					
N	13,423	3,431	503	4,760	6,618
Age	8.16 (2.74)	13.73 (2.40)	24.09 (6.18)	56.63 (7.62)	72.22 (4.74)
BMI	17.62 (3.15)	20.43 (4.01)	24.89 (5.93)	26.91 (5.01)	27.09 (4.43)
Valid wear time hours	22.12 (2.91)	21.16 (4.03)	21.65 (3.27)	22.61 (2.55)	23.78 (1.08)
PA intensity	-2.12 (0.19)	-2.20 (0.19)	-2.37 (0.20)	-2.49 (0.20)	-2.67 (0.22)
PA volume	43.23 (25.28)	42.28 (16.83)	40.39 (11.11)	39.58 (13.10)	31.23 (10.39)
Sleep duration	424.66 (126.66)	374.98 (131.77)	349.94 (103.36)	398.32 (69.36)	397.36 (74.35)
Sleep efficiency	0.78 (0.13)	0.80 (0.15)	0.87 (0.09)	0.87 (0.07)	0.86 (0.06)
Sleep onset	20.93 (1.21)	22.17 (2.03)	23.97 (1.65)	23.65 (1.35)	24.41 (1.32)
Sleep regularity	54.58 (12.72)	54.08 (13.27)	54.66 (13.11)	59.41 (11.59)	54.67 (12.59)
Accelerometer Wear Location					
Wrist	5,547 (41.32%)	2,110 (61.50%)	503 (100.00%)	4,760 (100.00%)	6,618 (100.00%)
Hip	7,876 (58.68%)	1,321 (38.50%)	-	-	-
Ethnicity					
Non-white	2,460 (18.33%)	492 (14.34%)	34 (6.76%)	91 (1.91%)	268 (4.05%)
Unclear	4,787 (35.66%)	1,912 (55.73%)	260 (51.69%)	1,360 (28.57%)	2,497 (37.73%)
White	6,176 (46.01%)	1,027 (29.93%)	209 (41.55%)	3,309 (69.52%)	3,853 (58.22%)
Region					
Europe	3,396 (25.30%)	1,239 (36.11%)	237 (47.12%)	4,345 (91.28%)	6,615 (99.95%)
Oceania	1,798 (13.39%)	694 (20.23%)	146 (29.03%)	415 (8.72%)	3 (0.05%)
South america	5,347 (39.83%)	1,144 (33.34%)	120 (23.86%)	-	-

Table 1 continued

Characteristic	Age group				
	2-11 years	12-18 years	19-35 years	36-65 years	66+ years
Africa	905 (6.74%)	174 (5.07%)	-	-	-
Asia	635 (4.73%)	95 (2.77%)	-	-	-
North america	1,342 (10.00%)	85 (2.48%)	-	-	-
Season					
Autumn	4,776 (35.58%)	821 (23.93%)	54 (10.74%)	1,164 (24.45%)	1,412 (21.34%)
Spring	2,620 (19.52%)	1,718 (50.07%)	285 (56.66%)	1,185 (24.89%)	2,149 (32.47%)
Summer	2,088 (15.56%)	372 (10.84%)	90 (17.89%)	1,200 (25.21%)	1,367 (20.66%)
Winter	3,939 (29.35%)	520 (15.16%)	74 (14.71%)	1,211 (25.44%)	1,690 (25.54%)
Sex					
Female	6,917 (51.53%)	1,757 (51.21%)	322 (64.02%)	2,671 (56.11%)	2,741 (41.42%)
Male	6,506 (48.47%)	1,674 (48.79%)	181 (35.98%)	2,089 (43.89%)	3,877 (58.58%)
Sleep Conditions Reported					
Yes	49 (0.37%)	1 (0.03%)	-	268 (5.63%)	690 (10.43%)
Socioeconomic Status					
Low	4,657 (34.69%)	898 (26.17%)	67 (13.32%)	1,180 (24.79%)	2,505 (37.85%)
Medium	4,532 (33.76%)	1,228 (35.79%)	278 (55.27%)	2,254 (47.35%)	2,592 (39.17%)
High	4,234 (31.54%)	1,305 (38.04%)	158 (31.41%)	1,326 (27.86%)	1,521 (22.98%)

Note. For categorical variables the value is the count, and percentage. For numeric variables the value is the Mean and

SD. N = 28,823

Table 2

Physical activity predicting sleep controlling for SES, sex, and BMI.

Term	Physical activity volume (z)				Physical activity intensity (z)			
	β [95% CI]	SE	t	p	β [95% CI]	SE	t	p
Sleep duration (z)								
(Intercept)	0.00 [-0.38, 0.39]	0.20	0.01	.991	0.01 [-0.35, 0.37]	0.18	0.05	.962
Physical activity	0.09 [-0.12, 0.29]	0.11	0.81	.422	0.09 [-0.01, 0.18]	0.05	1.76	.084
Age	0.00 [-0.01, 0.00]	0.00	-0.80	.425	0.00 [0.00, 0.00]	0.00	-0.87	.385
Physical activity ²	0.00 [-0.03, 0.02]	0.01	-0.36	.722	-0.01 [-0.04, 0.02]	0.02	-0.58	.562
Physical activity $\times$ age	0.00 [0.00, 0.00]	0.00	-0.44	.662	0.00 [0.00, 0.00]	0.00	-1.29	.202
Age $\times$ Physical activity ²	0.00 [0.00, 0.00]	0.00	-0.49	.625	0.00 [0.00, 0.00]	0.00	-0.43	.666
Sleep efficiency (z)								
(Intercept)	0.12 [-0.15, 0.39]	0.14	0.88	.381	0.15 [-0.11, 0.40]	0.13	1.13	.261
Physical activity	0.10 [-0.04, 0.25]	0.07	1.39	.172	0.01 [-0.03, 0.06]	0.02	0.53	.597
Age	0.00 [0.00, 0.00]	0.00	0.06	.952	0.00 [0.00, 0.00]	0.00	-0.32	.753
Physical activity ²	-0.01 [-0.03, 0.00]	0.01	-1.59	.117	0.01 [-0.01, 0.03]	0.01	0.73	.466
Physical activity $\times$ age	0.00 [0.00, 0.00]	0.00	-1.24	.222	0.00 [0.00, 0.00]	0.00	-1.04	.302
Age $\times$ Physical activity ²	0.00 [0.00, 0.00]	0.00	0.55	.585	0.00 [0.00, 0.00]	0.00	-1.03	.305
Sleep onset (z)								
(Intercept)	-0.10 [-0.38, 0.18]	0.14	-0.68	.500	-0.06 [-0.33, 0.21]	0.14	-0.44	.657
Physical activity	0.04 [-0.05, 0.14]	0.05	0.86	.392	-0.05 [-0.11, 0.00]	0.03	-1.84	.071
Age	0.00 [0.00, 0.00]	0.00	1.13	.261	0.00 [0.00, 0.00]	0.00	0.70	.486
Physical activity ²	-0.01 [-0.02, 0.00]	0.01	-1.88	.066	-0.01 [-0.03, 0.02]	0.01	-0.58	.566
Physical activity $\times$ age	0.00 [0.00, 0.00]	0.00	-2.52	.015	0.00 [0.00, 0.00]	0.00	-0.36	.723
Age $\times$ Physical activity ²	0.00 [0.00, 0.00]	0.00	2.47	.016	0.00 [0.00, 0.00]	0.00	1.20	.234
Sleep regularity (z)								

Table 2 continued

Term	Physical activity volume (z)				Physical activity intensity (z)					
	β	[95% CI]	SE	t	p	β	[95% CI]	SE	t	p
(Intercept)	0.35	[0.17, 0.52]	0.09	3.87	< .001	0.42	[0.23, 0.61]	0.10	4.33	< .001
Physical activity	0.14	[0.08, 0.20]	0.03	4.33	< .001	0.11	[0.05, 0.17]	0.03	3.61	< .001
Age	0.00	[0.00, 0.00]	0.00	0.51	.614	0.00	[0.00, 0.00]	0.00	-0.79	.432
Physical activity ²	-0.02	[-0.03, -0.01]	0.01	-3.14	.003	-0.02	[-0.05, 0.01]	0.02	-1.43	.158
Physical activity $\times$ age	0.00	[0.00, 0.00]	0.00	1.90	.062	0.00	[0.00, 0.00]	0.00	-3.27	.002
Age $\times$ Physical activity ²	0.00	[0.00, 0.00]	0.00	-4.10	< .001	0.00	[0.00, 0.00]	0.00	-0.52	.602

Note. Adjusted for SES, sex, and BMI. Outcomes variables are listed in the column headers.

Table 3

Sleep predicting physical activity controlling for SES, sex, and BMI

Term	Physical activity volume (z)				Physical activity intensity (z)			
	β [95% CI]	SE	t	p	β [95% CI]	SE	t	p
Sleep duration (Intercept)	1.15 [0.81, 1.49]	0.17	6.59	< .001	1.07 [0.82, 1.32]	0.13	8.38	< .001
Sleep duration	-0.01 [-0.12, 0.10]	0.06	-0.16	.874	0.02 [-0.01, 0.05]	0.01	1.22	.226
Age	-0.02 [-0.02, -0.01]	0.00	-7.52	< .001	-0.02 [-0.03, -0.01]	0.00	-6.68	< .001
Sleep duration ²	0.00 [-0.04, 0.03]	0.02	-0.30	.766	0.00 [-0.01, 0.01]	0.00	-0.17	.866
Sleep duration $\times$ age	0.00 [0.00, 0.00]	0.00	-0.08	.936	0.00 [0.00, 0.00]	0.00	0.59	.556
Age $\times$ Sleep duration ²	0.00 [0.00, 0.00]	0.00	-0.62	.540	0.00 [0.00, 0.00]	0.00	-0.19	.851
Sleep efficiency (Intercept)	1.13 [0.79, 1.48]	0.18	6.42	< .001	1.07 [0.82, 1.32]	0.13	8.39	< .001
Sleep efficiency	0.05 [-0.03, 0.12]	0.04	1.17	.247	0.02 [0.00, 0.04]	0.01	2.10	.039
Age	-0.02 [-0.02, -0.01]	0.00	-7.30	< .001	-0.02 [-0.03, -0.01]	0.00	-6.67	< .001
Sleep efficiency ²	0.01 [-0.01, 0.02]	0.01	1.26	.215	0.00 [0.00, 0.01]	0.00	1.61	.111
Sleep efficiency $\times$ age	0.00 [0.00, 0.00]	0.00	-0.82	.417	0.00 [0.00, 0.00]	0.00	-1.72	.090
Age $\times$ Sleep efficiency ²	0.00 [0.00, 0.00]	0.00	-0.56	.579	0.00 [0.00, 0.00]	0.00	-0.44	.664
Sleep onset (Intercept)	1.17 [0.82, 1.51]	0.18	6.62	< .001	1.08 [0.83, 1.33]	0.13	8.40	< .001
Sleep onset	0.03 [-0.08, 0.13]	0.05	0.50	.622	-0.01 [-0.03, 0.02]	0.01	-0.52	.608
Age	-0.02 [-0.02, -0.01]	0.00	-7.37	< .001	-0.02 [-0.03, -0.01]	0.00	-6.68	< .001
Sleep onset ²	-0.02 [-0.04, -0.01]	0.01	-2.57	.013	-0.01 [-0.01, 0.00]	0.00	-2.04	.044
Sleep onset $\times$ age	0.00 [0.00, 0.00]	0.00	-0.64	.523	0.00 [0.00, 0.00]	0.00	-0.46	.649
Age $\times$ Sleep onset ²	0.00 [0.00, 0.00]	0.00	1.30	.199	0.00 [0.00, 0.00]	0.00	0.71	.479
Sleep regularity (z)								

Table 3 continued

Term	Physical activity volume (z)				Physical activity intensity (z)					
	β	[95% CI]	SE	t	p	β	[95% CI]	SE	t	p
(Intercept)	1.10	[0.75, 1.44]	0.18	6.24	< .001	1.06	[0.81, 1.31]	0.13	8.40	< .001
Sleep regularity	0.12	[0.04, 0.20]	0.04	3.07	.003	0.05	[0.04, 0.07]	0.01	7.41	< .001
Age	-0.02	[-0.02, -0.01]	0.00	-7.16	< .001	-0.02	[-0.03, -0.01]	0.00	-6.67	< .001
Sleep regularity ²	0.01	[0.00, 0.03]	0.01	1.54	.129	0.00	[-0.01, 0.01]	0.00	-0.58	.563
Sleep regularity $\times$ age	0.00	[0.00, 0.00]	0.00	-2.00	.050	0.00	[0.00, 0.00]	0.00	-4.52	< .001
Age $\times$ Sleep regularity ²	0.00	[0.00, 0.00]	0.00	-1.25	.215	0.00	[0.00, 0.00]	0.00	-0.07	.945

Note. Adjusted for SES, sex, and BMI. Outcomes variables are listed in the row headers.

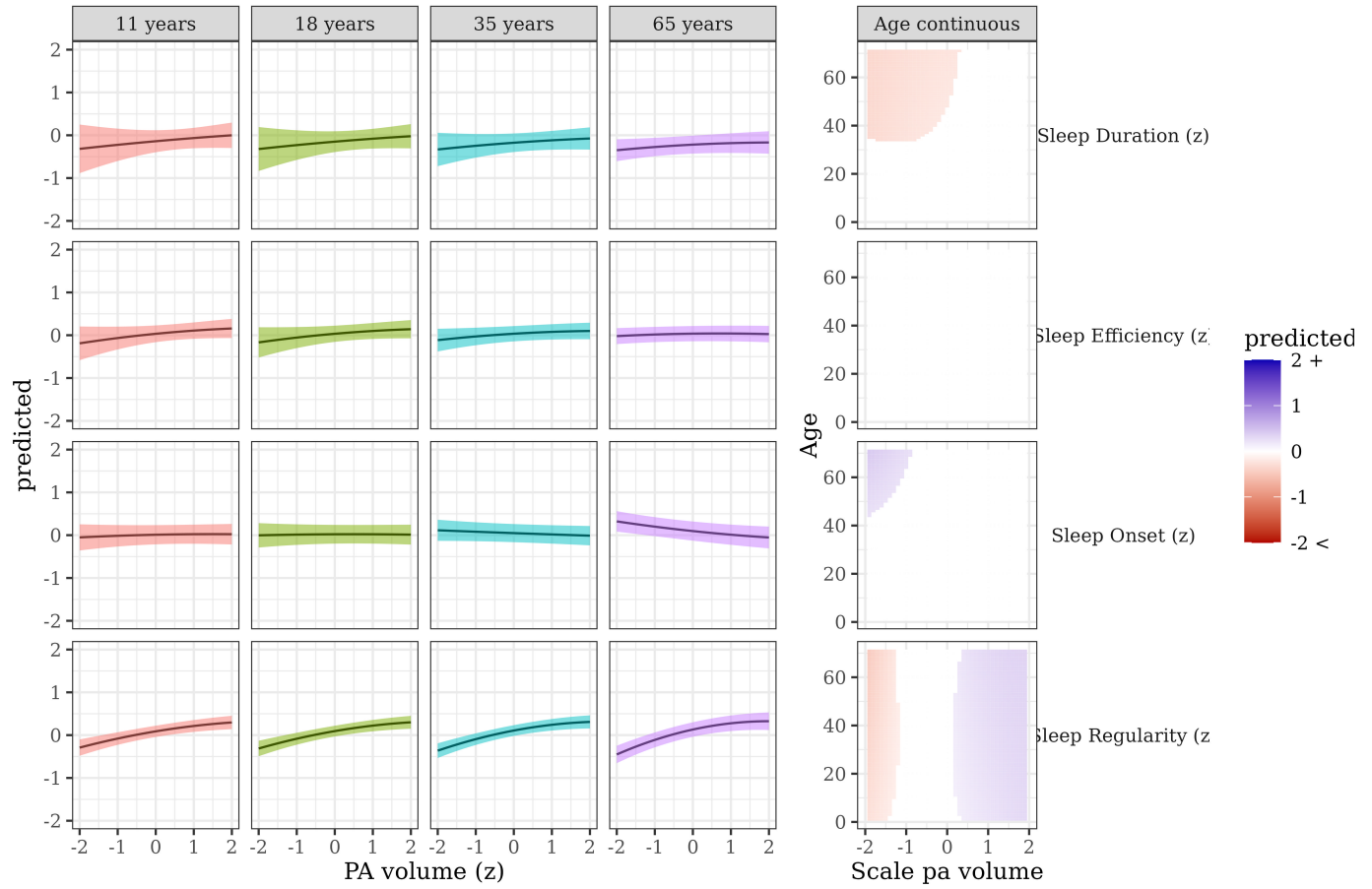


Figure 1. Sleep metrics on physical activity volume. The panels on the left show the curvilinear relationship between PA volume and each of the sleep outcomes at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero. These plots demonstrate exact turning points where the effects of PA volume change by age.

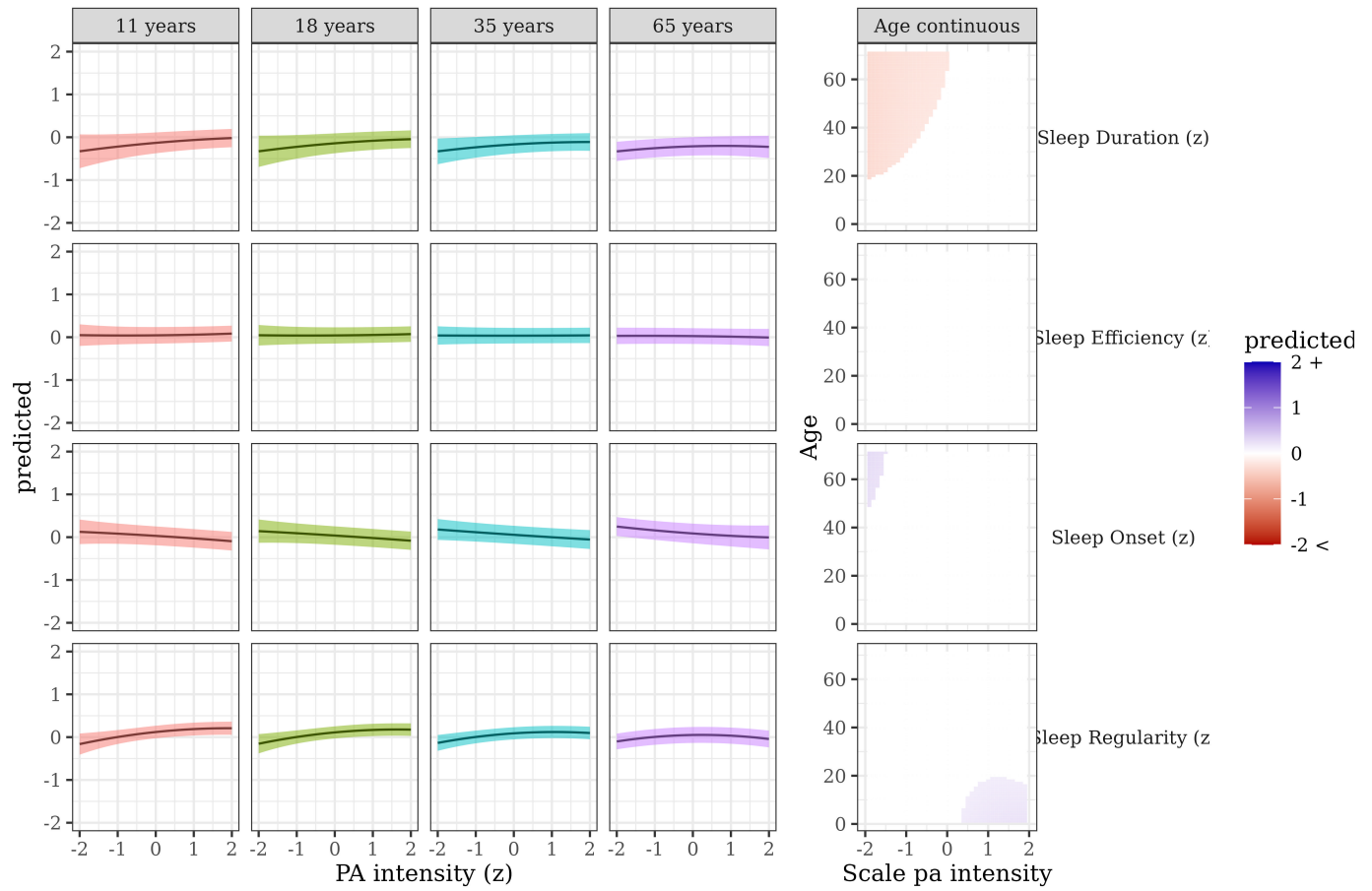


Figure 2. Sleep metrics on physical activity intensity. The panels on the left show the curvilinear relationship between PA intensity and each of the sleep outcomes at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero.

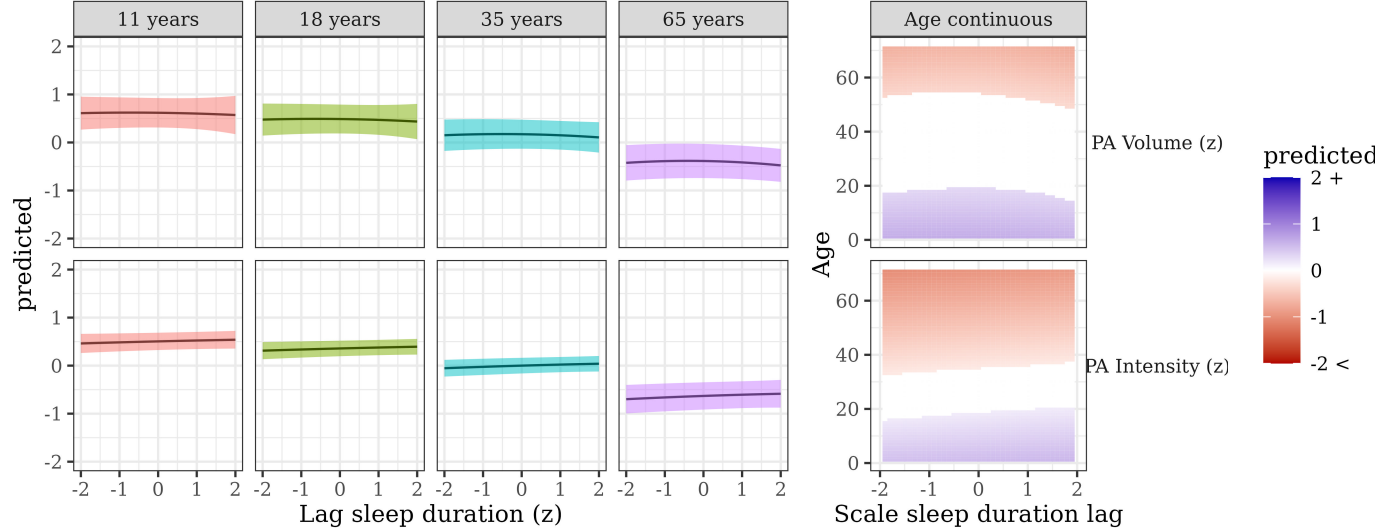


Figure 3. Physical activity by sleep duration. The panels on the left show the curvilinear relationship between sleep duration and each of the physical activity outcomes at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero.

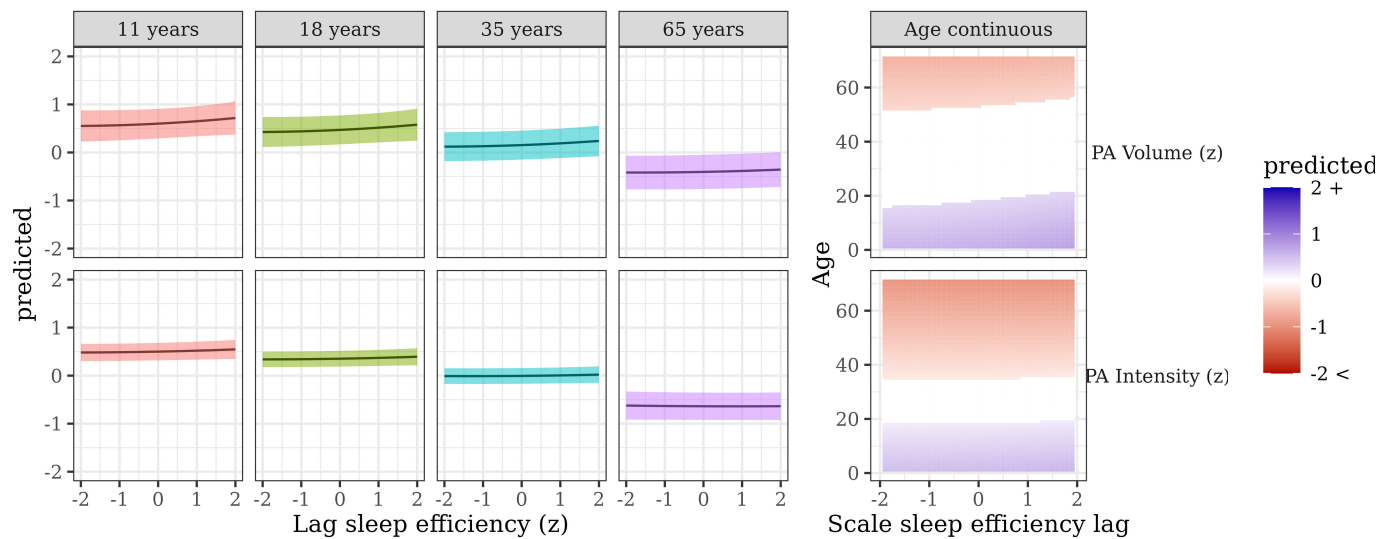


Figure 4. Physical activity by sleep efficiency. The panels on the left show the curvilinear relationship between sleep efficiency and each of the physical activity outcomes at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero.

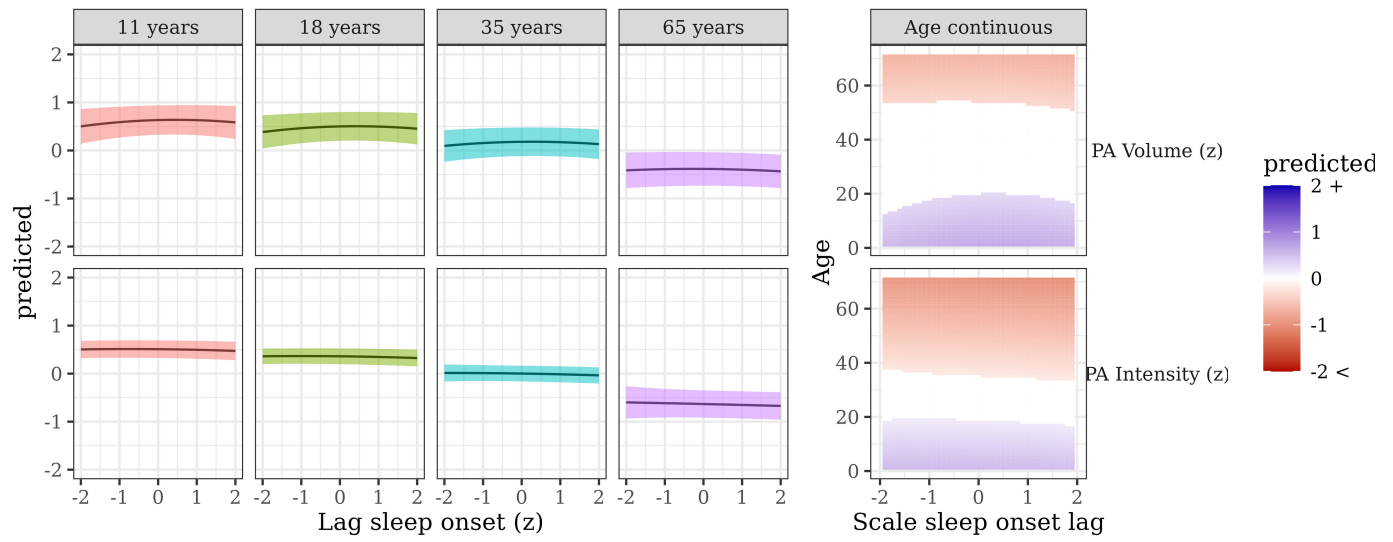


Figure 5. Physical activity by sleep onset. The panels on the left show the curvilinear relationship between sleep onset and each of the physical activity outcomes at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero.

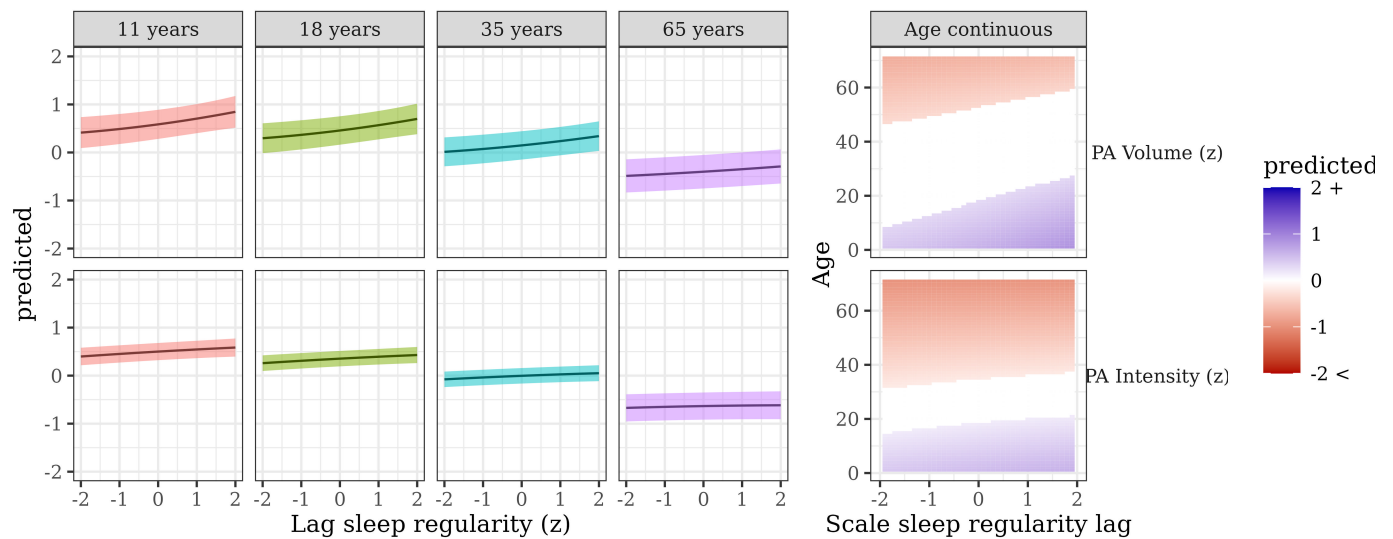


Figure 6. Physical activity by sleep regularity. The panels on the left show the curvilinear relationship between sleep regularity and each of the physical activity outcomes at the ages indicated in each column. The panels on the right show the same relationships but by age continuously. The white band indicates predictions which were not significantly different from zero.